

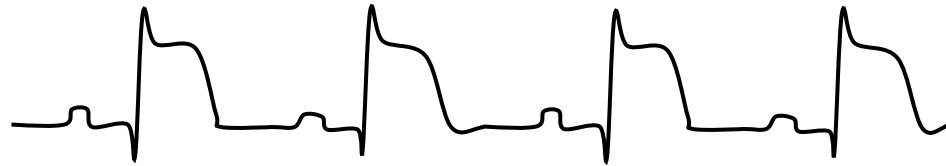
Early Tx

PCI

Late Tx

Cases

Management of Acute Coronary Syndromes



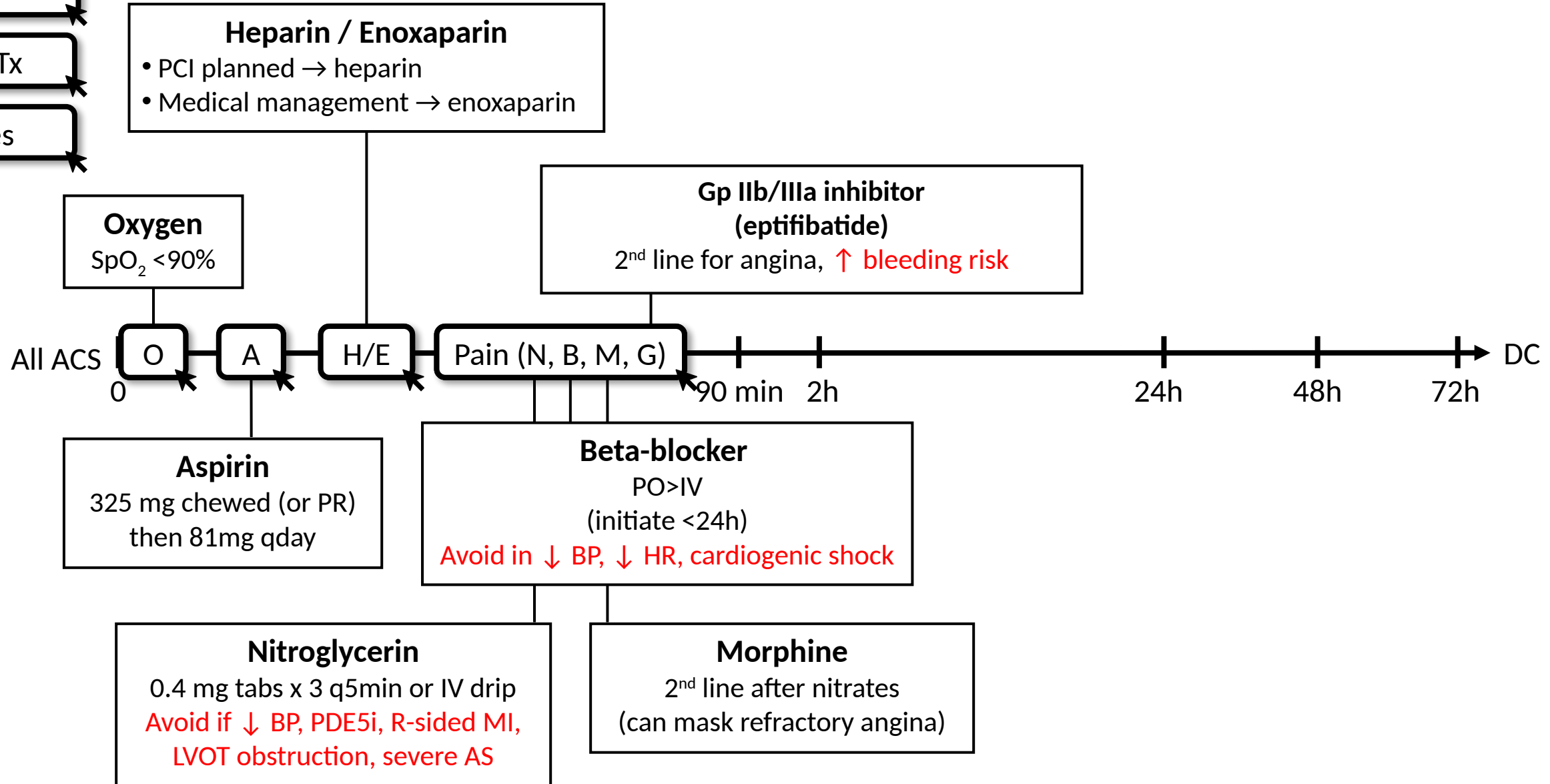
Early Tx

PCI

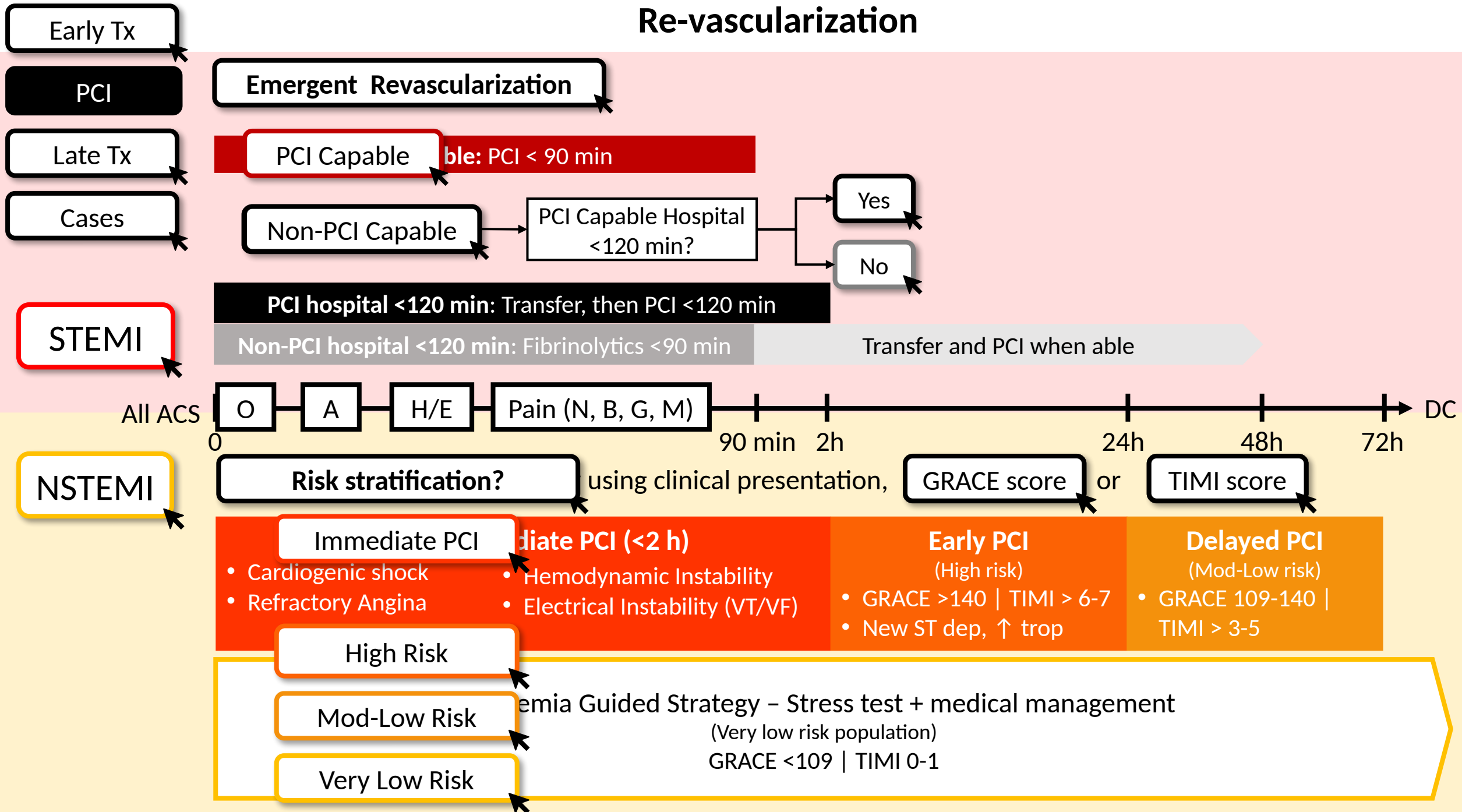
Late Tx

Cases

Early ACS Management - NSTEMI and STEMI



Re-vascularization



Late ACS Management – NSTEMI and STEMI

- Early Tx
- PCI
- Late Tx
- Cases

P₂Y₁₂ inhibitor
Initiate after PCI, ticagrelor is superior in STEMI
Avoid if anticipating CABG

Statin
(initiate <24h)
High dose, high intensity (atorvastatin, rosuvastatin)

ACE/ARB
(initiate before discharge)
• STEMI – all patients
• NSTEMI – LVEF <40% ± HTN, DM, CKD

STEMI

Emergent PCI



NSTEMI

Urgent/Immediate PCI **Early PCI** **Delayed PCI**

Beta-blocker
(initiate <24h)
Avoid in ↓ BP, ↓ HR, cardiogenic shock

MRA
(initiate before discharge)
LVEF <40%, HF, DM and tolerating other meds

Early Tx

PCI

Late Tx

Cases

1

2

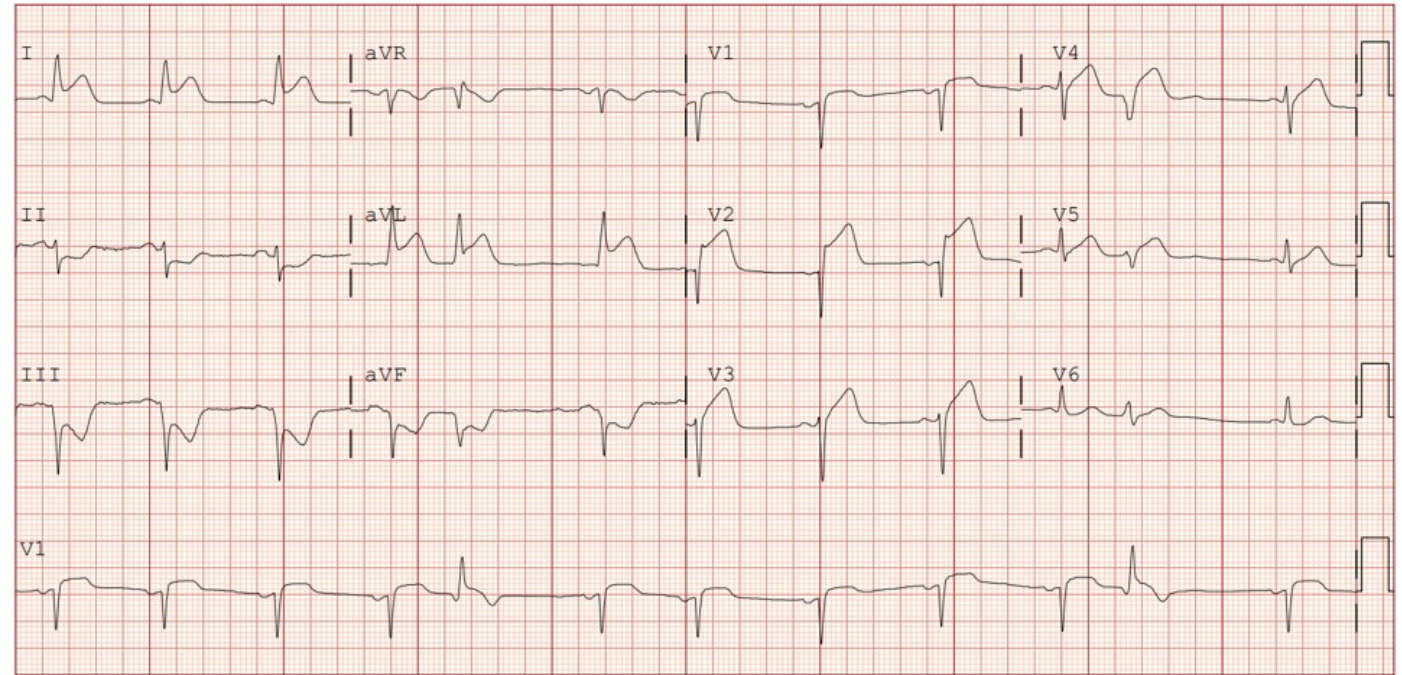
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Case 1

61-year-old with PMHx of hypertension, diabetes and hyperlipidemia presents with chest pain. Vitals show T 36.4C, HR 114, BP 105/78, SaO2 94%.

When should this patient undergo revascularization (if at all)?

EMERGENTLY! This ECG shows ST elevations in I, aVL, V1-V4 consistent with an **LAD STEMI**. At PCI capable hospital, the cath lab should be activated and **PCI performed <90 min.**



Courtesy of Benjamin Titus, MD.

What if the patient was at a non-PCI capable hospital?

Transfer to PCI-capable hospital if estimated door-to-balloon time is still **<120 min.**

If not, treat with fibrinolytics <90 min → transfer for delayed PCI <24 hours. Consider individual bleeding risk.

What medical therapy would you initiate while awaiting PCI?

- Aspirin 325 mg chewed (or PR)
- Heparin drip given planned PCI
- Hold on P₂Y₁₂ inhibitor in the event of needing a CABG
- PO beta-blocker (e.g., metoprolol) for angina
- PRN nitrates or morphine

They undergo PCI with DES to LAD. HR 80, BP 135/90. TTE shows EF of 50%. Labs show Cr 0.9, K 4.0

What medical therapy would you initiate prior to discharge?

- Aspirin + ticagrelor (superior P₂Y₁₂ inhibitor in STEMI)
- High intensity statin (e.g., rosuvastatin 20 mg)
- Beta-blocker
- ACE/ARB (for all STEMI)
- No indication for MRA (LVEF>40%)

Early Tx

PCI

Late Tx

Cases

1

2

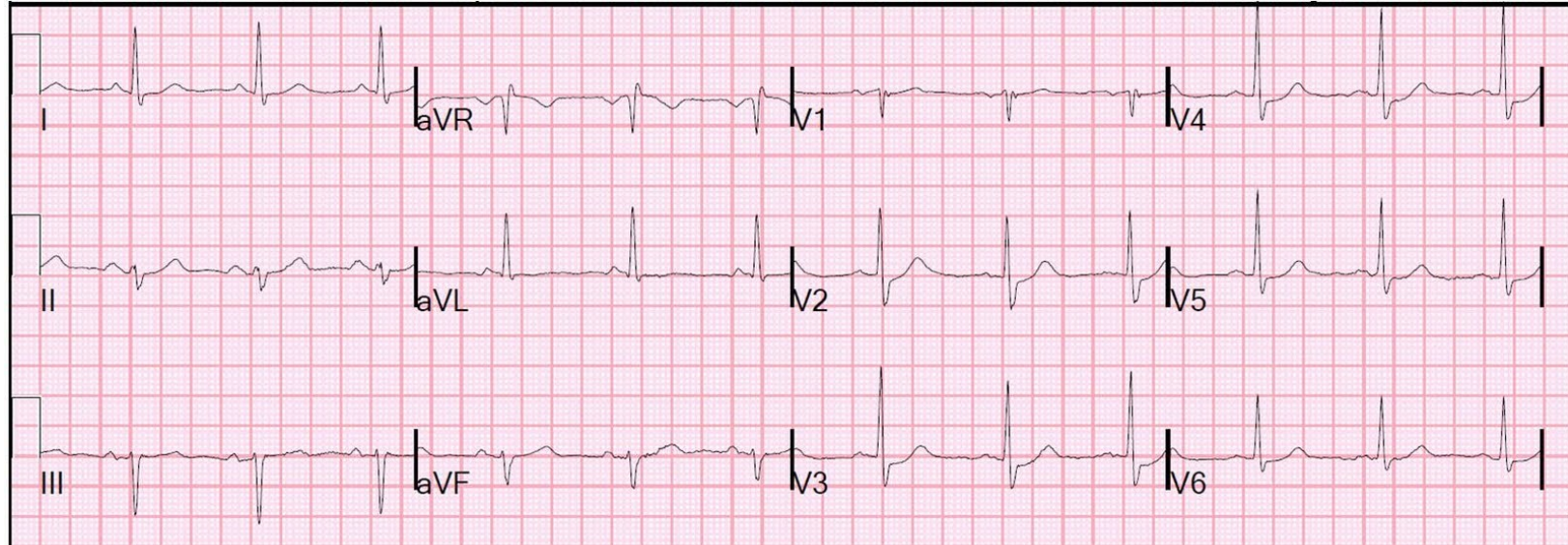
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Case 2

45-year-old with DM type II presents with crushing substernal chest pain. Vitals show HR is 110, BP 82/45, SaO2 82% on room air.

How would you manage pain?

May consider **GPIIb/IIIa inhibitor (eptifibatide)**.
Nitrates, morphine and beta-blockers are contraindicated with hypotension.



Courtesy of Stephen Smith, MD. [Dr. Smith's ECG blog](#). CC BY-NC 4.0

When should this patient undergo revascularization (if at all?)

Immediate/ urgent (<120 min). This is an **NSTEMI** (ST depressions V2-V4) but has high-risk features (cardiogenic shock, refractory angina since most anti-anginals are contraindicated). Other high-risk features that would prompt immediate PCI include ventricular arrhythmias, persistent angina

What medical therapy would you initiate while awaiting PCI?

- Oxygen (hypoxic on room air)
- Aspirin 325 mg chewed (or PR)
- Heparin drip given planned PCI
- Hold on P2Y12 inhibitor in the event of needing a CABG
- High intensity statin

They undergo PCI with DES to proximal LAD. VS are now HR 80, BP 135/90. TTE shows EF of 28%. Labs show Cr 0.9, K 4.0

What medical therapy would you initiate prior to discharge?

- Dual antiplatelet (e.g., aspirin + clopidogrel)
- Beta-blocker when euvolemic and no longer decompensated
- ACE/ARB (given DM and EF<40%)
- MRA (given DM and EF<40%)

Early Tx

PCI

Late Tx

Cases

1

2

3

Case 3

41-year-old with PMHx of anxiety with one episode of chest pain. Vital signs show HR 85, BP 145/85, SaO₂ 94% on room air.

ECG: T-wave inversions in precordial leads.

Labs: hs-troponin I is 25 ng/L → 110 ng/L (H) → 15 ng/L.

What is your diagnosis?

NSTEMI. This meets criteria with elevated troponin + chest pain + ECG change.

How would you estimate their mortality risk?

Very low risk. GRACE and TIMI score can be used to risk stratify patients with NSTEMI.

GRACE score is 53 → 0.8% probability of death in 6 months. TIMI score is 1 → 5% all cause mortality at 14 days.

What revascularization strategy would you choose?

Very low risk patients can be managed with an ischemia guided strategy. This includes a stress test and medical management, especially considered when patient preference.

While awaiting stress test, what therapy would you initiate?

- Aspirin
- Lovenox (superior to heparin in medical management)
- High intensity statin
- Beta blocker
- Nitro PRN chest pain
- P₂Y₁₂ inhibitor can be held prior to knowing coronary anatomy

A treadmill stress test (TTE) showed inducible wall motion abnormalities concerning for ischemia. TTE showed LVEF of 45%.

What is your next step in management?

PCI followed by medical management of NSTEMI.

2025 ACC/AHA ACS Guideline — What Changed Since This Talk (Oct 2023)

Antiplatelet therapy

- Ticagrelor OR prasugrel preferred over clopidogrel in STEMI & NSTEMI-ACS undergoing PCI (Class 1); clopidogrel if contraindicated/unavailable.
- DAPT ≥12 mo default if not high bleeding risk. To cut bleeding: PPI if GI-bleed risk; ticagrelor monotherapy ≥1 mo post-PCI if DAPT tolerated; if long-term OAC needed, stop aspirin 1–4 wk post-PCI, continue P2Y₁₂ (pref. clopidogrel).

Lipids

- High-intensity statin for ALL ACS, option of concurrent ezetimibe. Add nonstatin (ezetimibe, PCSK9 mAb, inclisiran, bempedoic acid) if LDL-C ≥70 mg/dL on max statin; intensifying reasonable if LDL-C 55–<70.

Revascularization & procedure

- NSTEMI-ACS intermediate/high risk: invasive approach during hospitalization. Low risk: routine invasive OR selective invasive.
- Radial access preferred over femoral (↓ bleeding, ↓ mortality); intravascular imaging for complex lesions (Class 1).
- Complete revascularization recommended in STEMI & NSTEMI-ACS; in cardiogenic shock, culprit-only PCI (no routine non-IRA PCI at same sitting); microaxial flow pump reasonable in select AMI-CS.

Other

- Transfusion to Hgb 10 g/dL may be reasonable in ACS with anemia (not actively bleeding). Lipid panel 4–8 wk after starting/changing therapy; refer to cardiac rehab.

Source: Rao SV et al. 2025 ACC/AHA/ACEP/NAEMSP/SCAI ACS Guideline. Circulation 2025;151. doi:10.1161/CIR.0000000000001309

Take-Home Points

- STEMI → emergent reperfusion: PCI <90 min if PCI-capable; transfer if door-to-balloon <120 min achievable; otherwise fibrinolytics <90 min then transfer for angiography (3–24 h).
- NSTEMI → risk stratify (GRACE/TIMI + clinical features): very high risk → immediate PCI (<2 h); high risk → early invasive (<24 h); mod-low → delayed invasive (25–72 h); very low risk → ischemia-guided strategy possible.
- Early meds for all ACS (no contraindications): aspirin 325 mg chewed, anticoagulation (heparin if PCI planned; enoxaparin if medical mgmt), nitrates, beta-blocker <24 h — never delay emergent revascularization for them.
- Within 24 h: high-intensity statin + P2Y12 inhibitor (post-PCI). Before discharge: ACEi/ARB (all STEMI; NSTEMI w/ EF<40%, HTN, DM, or CKD) and MRA (EF<40%, HF, or DM, if tolerating BB + ACEi/ARB).
- Avoid routine O2 if SpO2 ≥90%; morphine is last-line (masks refractory angina). Ongoing chest pain on max therapy = ongoing ischemia → revascularize.

Attribution & Changelog

Original

- "Management of Acute Coronary Syndrome" — Coffey Thielen S, Slaven S, Burke J; eds. Zhang Y, Fainstad B. TeachIM.org, Oct 2023. https://teachim.org/teaching_material/acs/ — CC BY-NC 4.0. Educational use only; not medical advice.

Corrections made to this copy (June 2026)

- Slide 2: oxygen indication "SpO2 >90%" → "SpO2 <90%" (per lesson handout table & 2025 guideline).
- Slide 7 (Case 3): hs-troponin units "μg/dL" → "ng/L" (3 instances).

Appended (originals untouched)

- 2025 ACC/AHA ACS guideline update slide; take-home points slide; this attribution slide. All animations and click-to-reveal builds preserved.

Companion files

- None — the lesson page's handout links mislink to Alcohol-related Liver Disease files; no ACS-specific handouts were available. Full optimized package: [ACS_Management_Lesson_Package.html](#).